



Simultaneous determination of eight tryptic peptides in musk using high-performance liquid chromatography coupled with tandem mass spectrometry

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ABSTRACT

In comparison of herbal medicines, less attention has been paid onto animal medicines, partially attributing to the protein-enriched property. Particularly, it is still challenging to conduct quality evaluation for the animal medicines because of the lack of a fit-for-purpose analytical tool. Herein, an attempt was made to propose a workflow allowing the quality assessment of animal medicines by LC-MS/MS, and musk that is one of the most precious traditional Chinese medicines was employed as a representative case for utility illustration. After the extraction of protein from musk with a well-defined protocol, tryptic digestion was conducted to hydrolyze proteins into peptides, and the peptide-enriched sample was subjected to nanoLC-Orbitrap MS measurement. The tandem mass spectral dataset was retrieved in Human Swiss-Prot FASTA database, and the sequences together with the sources of 733 tryptic peptides, in total, were annotated. Because of the abundant distributions, eight peptides were chosen as the analytes for quantitative measurements, and their quantitative MS parameters, such as ion transitions and collision energies, were rapid optimized in an authentic compound-free manner using online energy-resolved MS (ER-MS). On the other side, the annotated peptides were structurally consolidated via synthesizing reference peptides. When the synthetic peptides were applied for parameter optimization with the authentic compound-dependent manner, the values were almost identical with those from online ER-MS measurements. After being validated with diverse assays, the developed method was applied for the simultaneous determination of eight peptides in 28 batches of musk samples, including captive (C1–C18) and wild ones (W1–W10). Significant differences took place for the content patterns of concerned tryptic peptides between the captive and wild musk samples. Trace distributions occurred for DVDAAYMNK in most batches. Captive samples were rich of QSLEASLAETGR, TLLDIDNTR, and EVATNSELVQSGK, whereas wild samples were able to accumulate YLGYLEQLLR. Together, the present study provided a meaningful approach for the quality evaluation of musk, as well as other peptide-enriched animal medicines, even if the absences of authentic peptides.

1. Introduction

Although less items archived in classical Chinese Materia Medica than herbal medicines, animal medicines serve as an important portion for traditional Chinese medicine (TCM). Particularly, animal medicines are extremely popular in medicinal practices, and even more precious than herbal medicines. Small molecules usually dominate in herbal medicines and also serve as the determinant role for the efficacy. On the

other side, animal medicines, e.g., musk (*Shexiang* in Chinese) [1–3], Colla Corii Asini (*E'jiao* in Chinese) [4] and cubilose (*Yanwo* in Chinese, also known as edible bird's nest) [5], are usually rich of proteins. Due to the lack of fit-for-purpose analytical tools and authentic proteins, it is still a great challenging to achieve quality evaluation for those animal medicines.

When the proteins are hydrolyzed into peptides, it isn't a challenging task to assign satisfactory chromatographic behaviors for those peptides

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using routine RPLC. The proteomics workflow that usually employs LC coupled with Qtof-MS or Orbitrap-MS, enables transferring the tandem mass spectral information to peptides list, and afterwards, some accessible databases, fortunately, are able to restore the peptides into proteins. However, the proteomics technology is merely advantageous at relative quantitative analysis, and cannot meet the requirements of quality assessment because of the limited sensitivity and dynamic range. On the other side, selected reaction monitoring (SRM) that is the unique function of triple quadrupole-mass spectrometry (QqQ-MS) has been recommended as the golden standard for the quantitative analysis [6,7], and wide applications have been conducted for the determination of tryptic peptides in complicated matrices, including animal medicines [7–11]. Actually, the determinant step is the pursuit of optimal mass spectrometric parameters, such as collision energy (CE) and ion transition, for SRM. Skyline software [12,13] has been demonstrated as a viable tool to suggest SRM parameters on the basis of dataset acquired on Qtof-MS or Orbitrap-MS platform; however, this robust tool still cannot tailor parameters for any QqQ-MS, attributing to the inter-laboratory and inter-instrument variations. We have proposed [14–18], fortunately, a versatile strategy namely online energy-resolved MS (ER-MS) being able to gain superior parameters for each peptide-of-interest on any QqQ-MS platform. Here, the workflow for quality assessment of animal medicines through quantitative analysis of tryptic peptides is therefore constructed as follows: 1) proteome is characterized using nanoLC–Orbitrap MS by tightly following the well-defined shotgun proteomics protocols; 2) amino acid sequence annotation and source attribution are conducted for all detected peptides using the accessible databases; and 3) quantitative MS parameter optimizations is carried out for primary peptides-of-choice by online ER-MS. Afterwards, reference peptides can facilitate converting SRM responses to contents through constructing regressive calibration curves.

Musk is a widely favored animal medicine in TCM practices for centuries, and honored as a “first-class” variety in Shen Nong’s Chinese Materia Medica. It is derived from the dried secretions of *Moschus berezovskii* Flerov., *M. sifanicus* Przewalski. and *M. moschiferus* Linnaeus. [19]. Musk is frequently featured by the therapeutic benefits such as resuscitation, refreshment, activating blood circulation and dredging collaterals, detumescence and pain relief based on TCM theories. Moreover, musk also serves as the primary raw material for perfume in the cosmetic industry [20]. The chemical composition of musk is extremely complicated, including muscone and its analogues, steroids, and proteins. Muscone is regarded as the most important effective component in musk. Interestingly, the proteins in musk have drawn increasing concerns in recent years, because of the promising anti-inflammatory benefits [21–24]. Therefore, it is urgent to include proteins as the indicators for the in-depth quality evaluation of musk.

In current study, an attempt was made to develop a reliable LC–MS/MS method enabling simultaneous determination of primary tryptic peptides in musk, and subsequently to assess the content differences between captive and wild musk samples. The findings are envisioned to provide a feasible tool for protein-focused quality evaluation of musk as well as other animal medicines, even though the authentic peptides were not available.

2. Experimental

2.1. Chemicals and reagents

Bivalirudin that acted as the internal standard (IS) was purchased from Standard Biotech Co. Ltd (Shanghai, China). Authentic peptides, including DVDAAYMNK, TLLEGESR, QSLEASLAETEGR, TLLDIDNTR, VLDELTLTK, EVATNSELVQSGK and YLGYLEQLLR, were commercially synthesized by Yuanye Biotech Co., Ltd. (Shanghai, China). The purity of each standard was determined to be greater than 99% using LC–MS. A total of twenty-eight batches of musk samples, including eighteen captive ones (C1–C18) and ten wild ones (W1–W10), were provided by

Zhangzhou Pien-Tze-Huang Pharmaceutical Co., Ltd. (Zhangzhou, China). All captive ones were sampled from *Moschus berezovskii* in the farms belonging to Pien-Tze-Huang Pharmaceutical Co., Ltd.; however, the original sources of those wide samples are unavailable. All voucher specimens are deposited in Modern Research Center for Traditional Chinese Medicine, School of Chinese Materia Medica, Beijing University of Chinese Medicine (Beijing, China).

Trypsin was obtained from Promega (Fitchburg, WI). Ultrafiltration units with 10 kDa molecular weight cut-off (MWCO) were purchased from Sartorius (Goettingen, Germany). LC–MS grade acetonitrile (ACN) and formic acid were obtained commercially from Thermo-Fisher (Pittsburgh, PA, USA). The deionized water was prepared in house using a Milli-Q Integral water purification device (Millipore, MA, USA). The other chemicals, such as dichloromethane and ethanol, were of analytical grade and purchased from Beijing Chemical Works (Beijing, China).

2.2. Sample preparation for qualitative characterization

IS solution (10 µg/mL) was prepared by dissolving bivalirudin with 30% aqueous ACN.

A well-defined protocol was tightly followed to yield the protein extract. In brief, a 200 mg aliquot was sampled from each batch and thoroughly mixed to generate the homogenized sample. The homogenized sample was then suspended in 5 mL dichloromethane. Ultrasonication (230 V, Branson model 5 510, Danbury, CT, USA) was carried out in water bath (25 °C) for 30 min, and dichloromethane was afterwards added to compensate for the lost weight during the ultrasonication. After centrifugation at 12 000 rpm for 10 min, the supernatants were deserted. The residues were then processed with 5 mL dichloromethane for one more time. Afterwards, the dichloromethane was replaced with ethanol and repeated the above steps. The residues resulted from ethanol treatment was extracted with water for three times and the supernatants were combined. The resultant supernatants were dried to residues using a rotary evaporator, and the residues were then reconstituted to the concentration as 10 mg/mL using Milli-Q water. The resultant solution was fortified with tri-(β -chloroethyl)-phosphate (TCEP, final concentration as 10 mmol/L), and then incubated at 67 °C for 10 min. Iodoacetamide (final concentration as 15 mmol/L) was introduced for alkylation via lucifugal incubation for 30 min. The incubate was transferred into an ultrafiltration unit and centrifuged at 10 000 rpm for 30 min after adding 200 µL of 8 mol/L aqueous urea. The residues were afterwards washed with 200 µL of 50 mmol/L NH_4HCO_3 and centrifuged at 10 000 rpm for 30 min. The proteins in the ultrafiltration unit were transferred into another tube and underwent digestion for 16 h using trypsin with appropriate concentration. After centrifugation at 12 000 rpm for 30 min, the supernatants were collected and a 50 µL aliquot was subsequently diluted with equal volume of IS solution to yield the testing sample for peptidome characterization. All musk samples, including eighteen captive batches (C1–C18) and ten wild batches (W1–W10), were processed in parallel with the homogenized sample to yield testing samples for quantitative measurements. The sampling amount was enlarged to 2000 mg for certain batches to ensure the quantifiability of analytes-of-interest.

Moreover, the mixed standard solution was prepared by dissolving all authentic peptides with 30% aqueous ACN to appropriate content profile. The mixed standard solution was afterwards serially diluted, and each diluted solution was fortified with equal volume of IS solution, to generate a set of calibration samples, finally. Among them, three calibration samples with high, medium, and low concentration levels also served as the quality control (QC) samples for method validation.

2.3. Peptidome mapping with nanoLC–Orbitrap MS

NanoLC–LTQ Orbitrap Velos Pro MS (Thermo-Fisher Scientific, Waltham, MA) was deployed for the shotgun proteome data acquisition

Table 1

Chromatographic and mass spectrometric parameters of peptides and internal standard.

Analyte	Sequence	t_R (min)	Ion transitions (m/z)	CE (eV)
1	DVDAAYMNK	3.76	513.7 > 812.4	20.55
2	TLLEGEESR	4.12	517.3 > 819.4	23.92
3	QSLEASLAETEGR	9.17	695.9 > 662.3	34.35
4	TLLDIDNTR	9.75	530.8 > 733.4	25.33
5	VLDELTLTK	10.08	516.3 > 819.5	22.03
6	EVATNSELVQSGK	10.73	681.4 > 201.1	41.16
7	SLDLDSIAEVK	13.31	651.9 > 316.2	32.00
8	YLGYLEQLLR	13.65	634.4 > 991.6	28.55
IS	Bivalirudin	12.19	1091.4 > 356.4	71.30

of the homogenized sample. Mobile phase composed of 0.1% aqueous formic acid (A) and ACN containing 0.1% formic acid (B) was pumped into an EASY-Spray™ C₁₈ column (0.75 × 100 mm, 3 μm, Thermo-Fisher Scientific) with a gradient program as below: 0–65 min, 5–30% B; 65–75 min, 30–50% B; 75–85 min, 50–100% B, 85–90 min, 100%; and flow rate, 300 nL/min. Injection volume was set as 2.0 μL. LTQ Orbitrap Velos Pro MS was operated in a collision-induced dissociation *top*-12 mode under the synchronization of Xcalibur 2.1.0 software and LTQ Tune Plus 2.5.5 software (Thermo-Fisher Scientific). Full scan mode was applied at a resolution of 60 000 FWHM (mass range: m/z 350–2 000) and automatically triggered twelve data-dependent MS/MS scans at a resolution of 7500 FWHM. The normalized CE at 35% and the activation time at 10 ms were implemented for spectral recording. The dataset was processed by Proteome Discoverer v.1.4 software (Thermo-Fisher Scientific). The SEQUEST algorithm was applied for amino acid sequencing via retrieving in Human Swiss-Prot FASTA database with defaulted settings [25,26].

2.4. Quantitative analysis with RPLC–SRM measurement

All testing samples prepared above, including C1–C18 and W1–W10, were involved for quantitative evaluations. Eight primary tryptic peptides, such as DVDAAYMNK, TLLEGEESR, QSLEASLAETEGR, TLLDIDNTR, VLDELTLTK, EVATNSELVQSGK, SLDLDSIAEVK, and YLGYLEQLLR were selected as the analytes attributing to their abundant distributions in the homogenized sample. To extend the applicability towards conventional LC platforms, a Waters Acquity UPLC HSS T3 column (2.1 × 100 mm, 1.8 μm, Milford, MA) instead of nanoLC column was deployed for chromatographic separations. Mobile phase consisting of 0.1% aqueous formic acid (A) and ACN fortified with 0.1% formic acid (B) was delivered in gradient as follows: 0–5 min, 12–15% B; 5–10 min, 15–30% B; 10–15 min, 30–35% B; 15.1–28 min, 12% B, and flow rate, 0.2 mL/min. The auto-sampler chamber was maintained at 4 °C, and the column was maintained at 40 °C.

SRM ion transitions and CEs of all eight tryptic peptides were optimized using online ER-MS [18] in an authentic compound-independent manner. Briefly, the quasi-molecular ion of each analyte was paired with each primary fragment ion species in MS² spectrum to generate several ion transition candidates. Each candidate subsequently derived a set of so-called pseudo-ion transitions (PITs) that were able to circumvent the surveillance of Analyst™ software (Version 1.6.2) [27]. A series of progressive CEs (step-size as 1 eV) within appropriate range was assigned to each PITs set. Taking TLLDIDNTR for instance, doubly charged molecular ion ($[M+2H]^{2+}$) was detected at m/z 530.79, and the primary fragment ion species such as m/z 846.43 (y_7^+), 733.35 (y_6^+), and 618.32 (y_5^+) were then employed for ion pairing to yield three ion transition candidates such as m/z 530.8 > 846.4, 530.8 > 733.4, and 530.8 > 618.3. Three sets of PITs were subsequently defined, for instance m/z 530.8 > 846.400, 530.8 > 846.401, 530.8 > 846.402, and so on, derived from m/z 530.8 > 846.4, and each PIT received a step-wise CE, e.g., 5 eV for m/z 530.8 > 846.400, 6 eV for m/z 530.8 > 846.401, and so forth. All PITs sets were typed into the monitoring list of SRM. The relative response dataset of each ion transition candidate was afterwards imported into GraphPad Prism 5.0 software (Graphpad, San Diego, CA) to construct the Gaussian-shaped break-down graph for each ion transition candidate. The optimal CE corresponded to the vertex of each regressive Gaussian curve.

The well-aligned information of eight tryptic peptides, including identities, ion transitions, optimal CE, and t_R , was then imported into the SRM monitoring list (Table 1).

2.5. Method validation

Following method development, assays such as linearity, sensitivity, precision, and accuracy were carried out. All variations were determined using relative standard deviations (RSDs%).

2.5.1. Linearity and sensitivity

Serial calibration samples prepared above participated in linearity and sensitivity assays. The linearity of each analyte was assayed by building calibration curve with more than six concentration levels, and each concentration level was run in triplicate. Calibration curve construction was accomplished by plotting the peak area ratio of an analyte and IS against the concentration over the tested range. A 1/ x weighting function was applied to advance the linear regression if necessary. The lower limit of quantification (LLOQ) and limit of detection (LOD) corresponded to the concentration levels at signal-to-noise ratios (S/N) of approximately 10 and 3, respectively.

2.5.2. Precision

Precision assay was conducted by assessing intra- and inter- day variations using QC samples. For intra-day variability assessment, all QC

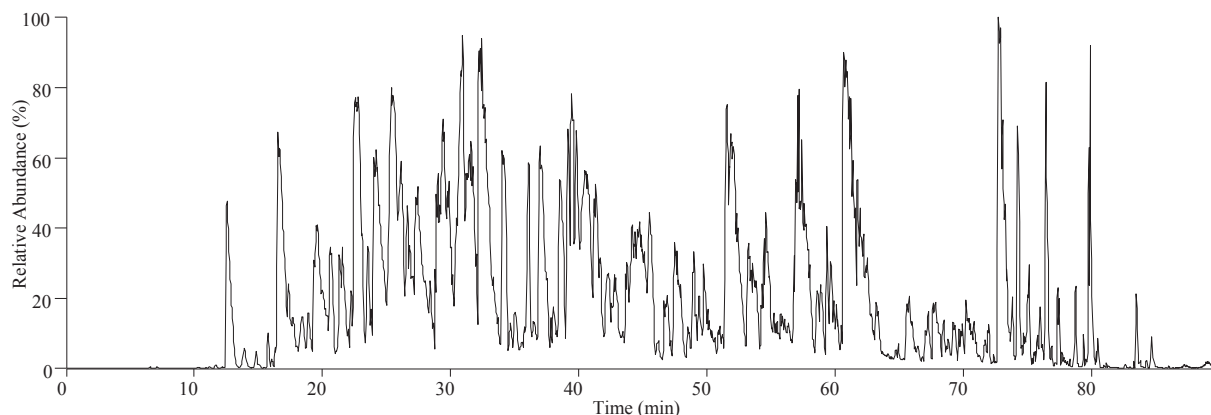


Fig. 1. Total ion current chromatogram of the homogenized sample using nanoLC coupled with LTQ Orbitrap Velos Pro MS.

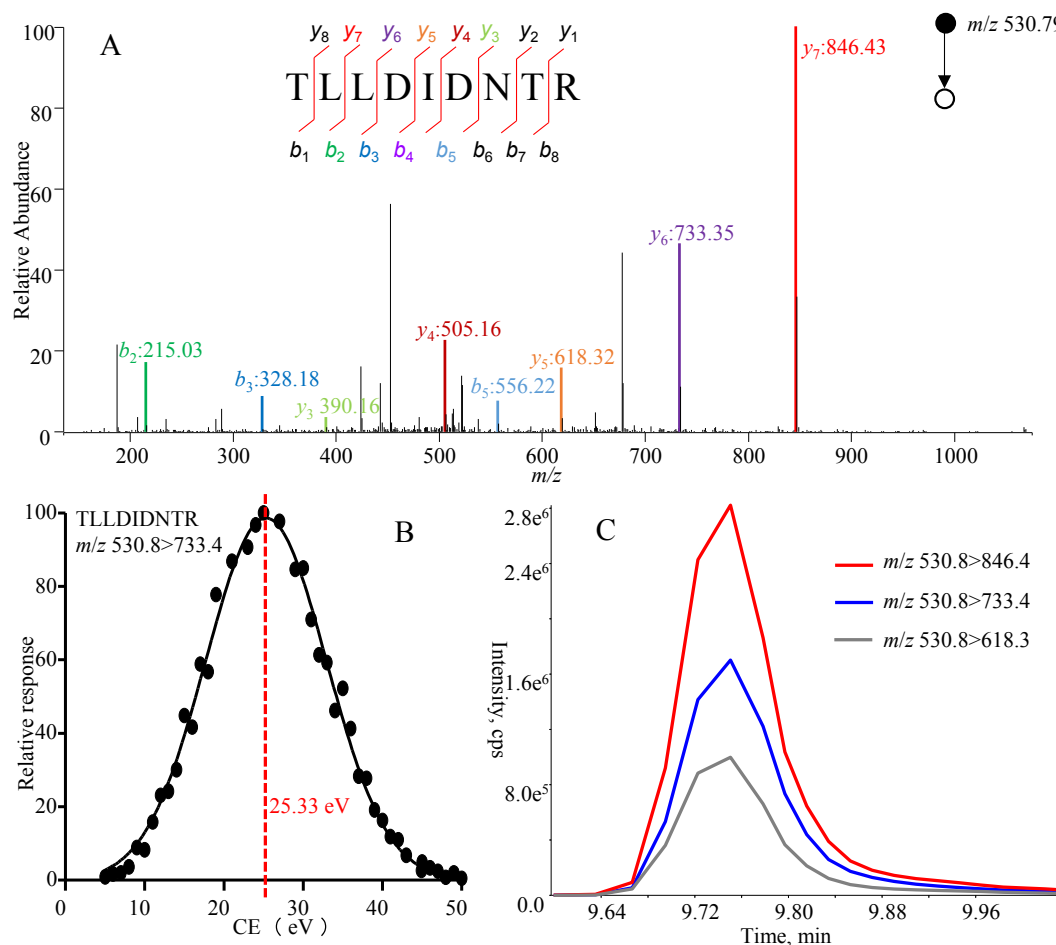


Fig. 2. Brief flow-chart for pursuing optimal ion transitions along with collision energies for tryptic peptides. The precursor ions as well as fragment ion species are provided by nanoLC coupled to LTQ Orbitrap Velos Pro MS, and structural annotation is completed through retrieving in Swiss-Prot database (A); the optimal CE of m/z 530.8 > 733.4 is yielded as 25.33 eV according to Gaussian fitting the serial responses of pseudo-ion transitions such as m/z 530.8 > 733.401, 530.8 > 733.402, 530.8 > 733.403, and so on, with Gaussian-shaped curve ($Y = 98.69 \times \exp(-0.5 \times ((X - 25.33)/7.61)^2)$) (B); and three ion transition candidates are constructed through pairing the precursor and fragment ion species, and m/z 530.8 > 846.4 receives the greatest responses, followed by m/z 530.8 > 733.4 and 530.8 > 618.3, sequentially (C).

samples were measured for six replicates within a single day, whereas all QC samples were examined in triplicate per day for three consecutive days to accomplish inter-day precision assay.

2.5.3. Repeatability and stability

To evaluate the repeatability, a selected sample was run for five replicates using the newly developed method. Moreover, the sample was deposited at 4 °C and analyzed at 0, 2, 4, 8, 12, 18, and 24 h to evaluate the stability.

2.5.4. Recovery

The recovery was deployed to evaluate the accuracy. Known amounts of standards solution (low, medium, and high levels) were added into a certain amount (50 mg) of C1 sample. The mixture was extracted following procedure described in Section 2.2 and then measured using the developed method.

3. Results and discussion

3.1. Peptidome characterization of musk

After being extracted from the homogenized sample, the proteins were hydrolyzed to peptides using trypsin, and peptidome was characterized using LC coupled to LTQ Orbitrap Velos Pro MS. The

chromatogram is illustrated in Fig. 1. Overall, satisfactory chromatographic pattern was yielded from nanoLC, and obviously, a number of signals occurred in the chromatogram. Afterwards, the dataset was processed with Proteome Discoverer software, and Swiss-Prot FASTA database was utilized for sequence annotation and source attribution. As a result, 733 tryptic peptides, in total, were detected and annotated in the peptidome dataset, and the information is summarized in Table S1 (Supplemental information A). A total of 79 proteins (Table S2, Supplemental information B) were responsible for the generation of those 733 peptide sequences after in-depth database retrieval. Particularly, keratins were the source of a large portion of annotated sequences, and serum albumin also played an important role for the peptidome.

Following sequence annotation, attention was subsequently paid onto selecting peptides as the analytes for quantitative measurements, and those ones with great sequence annotation confidences as well as significant occurrences were favored as the quality indicators for musk. Eight tryptic peptides, including DVDAAYMNK, TLLEGEESR, QSLEASLAETGR, TLLDIDNTR, VLDELTLTK, EVATNSLVQSGK, SLDLDSIIAEVK, and YLGYLEQLLR were ultimately chosen as analytes because of their dramatic intensities as well as significant matching rates between the multi-stage mass spectral profiles and the information archived in the databases.

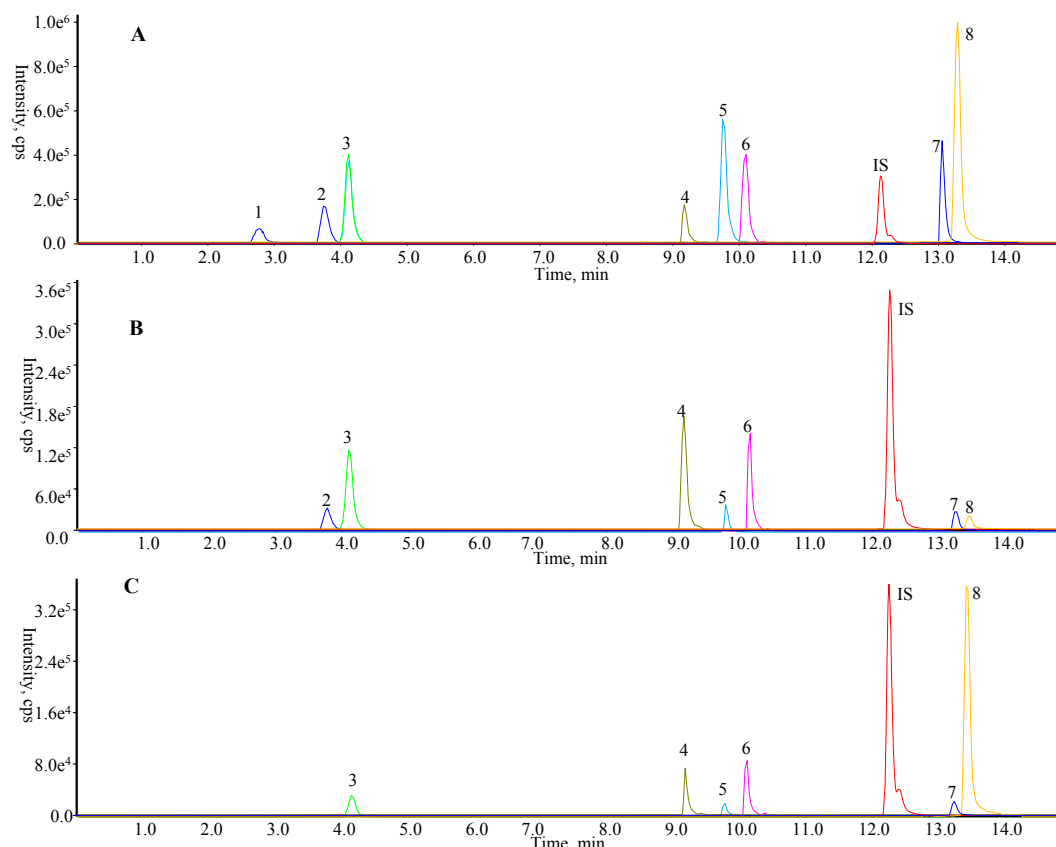


Fig. 3. Representative overlaid extracted ion current chromatograms of the mixed standard solution (A), captive musk sample (B), and wild musk sample (C). 1: DVDAAYMNK; 2: TLLEGEESR; 3: QSLEASLAETEGR; 4: TLLDIDNTR; 5: VLDELTLTK; 6: EVATNSELVQSGK; 7: SLDLDSIIAEVK; and 8: YLGYLEQLLR.

3.2. Method development and validation

To consolidate the sequences for analytes-of-choice, the sequences annotated by database retrieval were manually synthesized, and the authentic peptides, the structures of which were confirmed with LC-IT-TOF-MS (Shimadzu) and ^1H NMR (Varian UNITY plus 500 MHz spectrometer, Varian Inc., Palo Alto, CA) measurements, were implemented for retention times and multi-stage mass spectra matching. As expected, great coincidence occurred between each pair of signals, suggesting that the selected peptides definitely exhibited the amino acid sequences as DVDAAYMNK, TLLEGEESR, QSLEASLAETEGR, TLLDIDNTR, VLDELTLTK, EVATNSELVQSGK, SLDLDSIIAEVK and YLGYLEQLLR, accordingly. Among them, seven ones were originated from keratins, whilst the other one namely YLGYLEQLLR was sourced from Alpha-S1-casein.

After identity confirmation, concerns were subsequently switched to transmit the qualitative information into quantitative MS parameters. In general, CE along with ion transition plays the determinant role for the peptide response [28–30] under SRM mode. Although Skyline software [31–33] is able to recommend SRM parameters, those parameters usually suffered from limited sensitivity. On the other side, the well-defined manual tuning protocol required authentic peptides; however, it was high cost to synthesize all peptides-of-choice and the manual tuning procedure was also time-consuming. Consequently, online ER-MS was introduced here to online optimize the parameters in an authentic compound-free manner.

TLLDIDNTR for instance, doubly charged protonated molecular ion ($[\text{M}+2\text{H}]^{2+}$) was detected at m/z 530.79, and then a set of singly charged fragment ion species such as m/z 846.43 (y_7^+), 733.35 (y_6^+), 618.32 (y_5^+), 556.22 (b_5^+), 505.16 (y_4^+), 390.16 (y_3^+), 328.18 (b_3^+), and 215.03 (b_2^+), corresponding to the dissociation of the amido bonds, were

observed in MS² spectrum (Fig. 2A). Because of the significant intensities, fragment ions including m/z 846.4, 733.4, and 618.3 were utilized to pair with the precursor ion (m/z 530.8) to generate three ion transition candidates, such as m/z 530.8 > 846.4, 530.8 > 733.4, and 530.8 > 618.3, and afterwards, three sets of PITs (i.e. m/z 530.800 > 846.401, 530.800 > 846.402, 530.800 > 846.403, etc.; m/z 530.800 > 733.401, 530.800 > 733.402, 530.800 > 733.403, etc.; m/z 530.800 > 618.301, 530.800 > 618.302, 530.800 > 618.303, etc.), were exactly derived from the three candidates. Regarding each PIT set, staggered CEs (step-size as 1 eV) amongst appropriate range were individually assigned to each PIT.

Because the quadrupole mass analyzer cannot recognize, actually, the slight differences (<50 mDa) among PITs when unit resolution is defined for either Q1 or Q3 cell, the response variations within each PIT set are solely initiated by CEs, and optimal CE thereafter corresponds to the greatest response. Moreover, the superior ion transition should be able to generate greater response at optimal CE than other candidates. After importing the relative response dataset into GraphPad Prism software, the breakdown graph of m/z 530.8 > 733.4 came out as Fig. 2B after Gaussian fitting and the optimal CE was calculated as 25.33 eV. Moreover, significant differences were observed among those ion transitions candidates, and m/z 530.8 > 846.4 received, obviously, the greatest responses, followed by m/z 530.8 > 733.4 and 530.8 > 618.3, sequentially (Fig. 2C).

Furthermore, the optimal parameters of the other tryptic peptides, including DVDAAYMNK, TLLEGEESR, QSLEASLAETEGR, VLDELTLTK, EVATNSELVQSGK, SLDLDSIIAEVK and YLGYLEQLLR, were also harvested in this manner. The Gaussian-shaped breakdown graphs of their superior ion transitions, such as m/z 513.7 > 812.4 (DVDAAYMNK), m/z 516.3 > 819.5 (VLDELTLTK), m/z 695.9 > 662.3 (QSLEASLAETEGR), m/z 634.4 > 991.6 (YLGYLEQLLR), m/z 517.3 > 819.4 (TLLEGEESR),

Table 2
Results of linear regression data, lower limits of quantification (LLOQs), limits of detections (LODs), intra- and inter-day variations, repeatability, stability, and recovery assays for all peptides.

Analyte	Calibration curves	r	Linear range (ng/mL)	LLOQs (ng/mL)	LODs (ng/mL)	Intra-day RSD (%)			Inter-day RSD (%)			Repeatability RSD (%)		Stability RSD (%)			Recovery (%)		
						Low	Medium	High	Low	Medium	High	Low	High	Low	Medium	High	Low	Medium	High
1	$y = 0.000157x - 0.000175$	0.9991	2.44–1250	0.61	0.20	5.93	4.28	4.69	7.69	7.80	7.38	N.A.	8.21	89.1	93.8	94.1	87.7	108.4	111.8
2	$y = 0.0014x - 0.00182$	0.9964	2.44–1250	2.44	1.22	5.44	5.26	4.56	10.63	7.49	9.45	9.30	8.00	87.3	110.2	88.5	111.9	90.3	110.6
3	$y = 0.00393x + 0.00125$	0.9992	2.44–1250	0.61	0.15	6.45	4.27	0.77	9.84	5.63	5.64	8.43	5.67	109.8	90.7	109.8	112.0	88.4	89.2
4	$y = 0.000915x + 0.000568$	0.9990	0.61–1250	0.61	0.15	6.06	8.49	1.07	10.19	6.62	3.57	5.67	5.55	111.9	90.3	110.6	109.8	88.4	112.0
5	$y = 0.00429x + 0.0186$	0.9996	0.61–1250	0.61	0.15	7.06	6.09	2.47	9.83	5.48	5.87	9.84	7.74	109.8	88.4	112.0	112.5	91.3	90.1
6	$y = 0.00829x + 0.0505$	0.9988	2.44–1250	2.44	0.61	5.85	5.86	3.40	8.74	7.13	4.94	4.90	7.28	90.7	109.8	88.4	112.0	88.4	89.2
7	$y = 0.00264x - 0.00364$	0.9991	0.423–866	0.42	0.11	6.30	6.26	1.88	7.64	8.04	5.49	11.65	7.36	112.0	93.2	89.2	112.0	93.2	89.2
8	$y = 0.00774x + 0.0132$	0.9991	0.61–1250	0.31	0.10	6.52	4.63	2.39	8.00	7.87	5.69	11.00	7.51	112.0	93.2	89.2	112.0	93.2	89.2

N.A., not applicable.

m/z 681.4 > 201.1 (EVATNSELVQSGK), and m/z 651.9 > 316.2 (SLDLSIIAEVK) are illustrated in Fig. S1 (Supplemental information A), and their optimal CE values were calculated as 20.55, 22.03, 34.35, 28.55, 23.92, 41.16, and 32.00 eV, respectively. It was worthwhile to mention that great fluctuation was observed for optimal CE values of the concerned peptides even though comparable dissociation pathways occurred among them; hence, it was of great importance to subtly optimize CE for each analyte.

The well-aligned information of eight tryptic peptides, including identities, ion pairs, and optimal CE values were then imported into the monitoring list of SRM. After that, the gradient elution program for RPLC was carefully evaluated. The combination of H₂O and ACN was found to be superior to the other available choices and the fortification of formic acid into H₂O as the additive was able to advance the overall chromatographic profile. The optimized elution program depicted in “2.4 Quantitative analysis with RPLC–SRM measurement” section enabled satisfactory retention for tryptic peptides. The tryptic peptides were observed amongst the retention time window as 3.76–13.65 min (Fig. 3A), and great resolution pattern occurred for all eight tryptic peptides.

3.3. Method validation

Following method development, assays in terms of linearity, sensitivity, precision, and accuracy were subsequently conducted by following the descriptions in the literature [34,35].

After measuring those calibration samples, the correlation coefficients (r) for all regressive curves were greater than 0.9964 (Table 2). Noteworthy, inferior ion transition as m/z 530.8 > 733.4 rather than the most sensitive one as m/z 530.8 > 846.4 was assigned to TLLDIDNTR because the upper limit of quantification of m/z 530.8 > 733.4 instead of m/z 530.8 > 846.4 was able to universally cover the significant distributions of TLLDIDNTR in all samples. LODs and LLOQs of all tryptic peptides were lower than 1.22 ng/mL and 2.44 ng/mL, respectively, suggesting that the newly developed method enabled sensitive measurements (Table 2). RSDs% that came from intra-day (0.77–8.49%), inter-day (3.57–10.63%), repeatability (<11.65%), and stability (5.55–8.21%) assays, were all lower than 15% (Table 2). Moreover, the recovery rates of all targeted peptides ranged from 87.3% to 112.5%. Above all, the developed method could fully fulfill the analytical demands of simultaneous determination of those eight tryptic peptides in musk samples.

Skyline software was widely popular due to the unique ability to recommend ion transitions as well as CE values for any concerned peptide after enquiring amino acid sequence [32–34]. Herein, comparison was carried out between those values suggested by Skyline software and the CE values optimized by online ER-MS. DVDAAYMNK for instance, optimal CE for m/z 513.7 > 812.4 was gained as 20.55 eV by online ER-MS, whereas Skyline suggested 28.10 eV to be the optimal value. Regarding to the breakdown graph of m/z 513.7 > 812.4 (Fig. S1), SRM response at 20.55 eV was two folds greater than the one corresponding to 28.10 eV. The comparisons of the other peptides also agreed that optimal CE values generated from online ER-MS were able to give greater SRM responses, usually more than two folds over the responses yielded by applying the values coming from Skyline software. Moreover, those authentic peptides were utilized for manual tuning of mass spectrometric parameters using the well-defined protocols [36], and the Gaussian-shaped breakdown graphs of targeted ion transitions, including m/z 513.7 > 812.4 for DVDAAYMNK, m/z 516.3 > 819.5 for VLDELTLTK, m/z 695.9 > 662.3 for QSLEASLAETEGR, m/z 634.4 > 991.6 for YLGYLEQLLR, m/z 517.3 > 819.4 for TLLEGESR, m/z 530.8 > 733.4 for TLLDIDNTR, m/z 681.4 > 201.1 for EVATNSELVQSGK, and m/z 651.9 > 316.2 for SLDLSIIAEVK, are illustrated in Fig. S2 (Supplemental information B). Overall, merely mild differences (Δ OCE < ± 0.3 eV) occurred for each pair of OCE values. Hence, the values generated from manual tuning with syringe pump infusion manner agreed well (tolerance as ± 0.3 eV

Table 3

The contents ($\mu\text{g/g}$) of peptides in 28 batches of musk samples (C1–C18 & W1–W10).

Batch No.	1	2	3	4	5	6	7	8
C1	N.D.	0.54	2.58	1.79	0.16	1.54	0.16	0.04
C2	N.D.	0.51	3.25	3.85	0.06	3.43	0.08	0.04
C3	0.20	0.33	2.29	1.48	0.19	0.97	0.41	0.04
C4	0.39	2.20	17.6	24.77	0.11	16.63	1.08	0.03
C5	0.24	N.D.	0.56	2.74	N.Q.	4.37	0.01	0.02
C6	N.D.	0.19	2.59	1.90	0.12	1.48	0.15	0.06
C7	N.D.	0.13	1.22	0.54	0.04	0.41	0.06	0.03
C8	N.D.	0.09	0.79	0.51	0.15	0.25	0.24	0.03
C9	N.D.	0.60	13.83	14.73	0.08	8.12	0.14	0.05
C10	N.D.	0.39	3.10	4.82	0.14	3.46	0.22	0.03
C11	N.D.	0.09	2.89	1.77	0.17	1.55	0.16	0.08
C12	N.D.	0.39	4.82	3.59	0.07	2.78	0.13	0.09
C13	0.05	0.43	9.73	6.40	0.16	5.11	0.25	0.03
C14	0.43	0.95	6.61	8.08	0.18	7.72	0.24	0.04
C15	0.11	2.55	20.89	24.47	0.24	14.18	0.38	0.03
C16	N.D.	0.38	4.12	2.46	0.31	2.20	0.59	0.08
C17	N.D.	0.35	0.72	0.34	0.14	0.27	0.18	0.05
C18	N.D.	1.78	2.45	1.28	0.19	1.25	0.20	0.06
W1	N.D.	N.D.	1.59	0.41	0.07	0.54	0.15	11.12
W2	N.D.	N.D.	0.70	0.36	0.14	0.28	0.11	17.82
W3	N.D.	N.D.	0.66	0.27	0.09	0.17	0.16	5.05
W4	0.16	N.D.	3.45	1.10	0.16	1.60	0.29	2.25
W5	0.16	N.D.	0.74	0.37	0.13	0.33	0.28	2.94
W6	N.D.	N.D.	0.38	0.34	0.09	0.22	0.10	0.52
W7	0.61	N.D.	1.03	0.42	0.12	0.42	0.15	10.78
W8	N.D.	N.D.	0.35	0.16	0.14	N.Q.	0.13	5.11
W9	2.72	N.D.	0.89	0.36	0.15	0.15	0.25	27.17
W10	0.82	N.D.	1.46	0.46	0.13	0.46	0.25	19.65

N.D., not detectable.

N.Q., not quantifiable.

[37]) with those ones from online ER-MS, suggesting the reliability of online ER-MS towards parameter optimization.

3.4. Variation assessment between captive and wild musk samples

Afterwards, those testing samples (C1–C18 & W1–W10) were measured, and the quantitative responses of all tryptic peptides were converted to contents using the regressive calibration curves (Table 2). The representative chromatogram is shown in Fig. 3. All quantitative results are summarized in Table 3. Overall, significant variations occurred for the contents of eight tryptic peptides in captive and wild musk samples, and great content span was observed for those tryptic peptides within either group. Except two batches (0.24 $\mu\text{g/g}$ in 11C-08 and 0.31 $\mu\text{g/g}$ in JY01), comparable content profiles were observed for VLDELTLTK between captive samples ($<0.19 \mu\text{g/g}$) and wild samples (0.07–0.16 $\mu\text{g/g}$). Similarly, except two batches (1.08 $\mu\text{g/g}$ in 09C-05 and 0.59 $\mu\text{g/g}$ in JY01), the contents of SLDLDSIAEVK in captive samples (0.01–0.41 $\mu\text{g/g}$) were comparable with the content pattern in wild samples (0.10–0.29 $\mu\text{g/g}$). It was worthy to underline that the contents of TLLEGEESR ($<2.55 \mu\text{g/g}$ in captive samples vs. not detectable in wild samples), QSLEASLAETEGR (0.56–20.89 $\mu\text{g/g}$ in captive samples vs. 0.35–3.45 $\mu\text{g/g}$ in wild samples), TLLDIDNTR (0.34–24.77 $\mu\text{g/g}$ in captive samples vs. 0.16–1.10 $\mu\text{g/g}$ in wild samples), and EVATNSSELVQSGK (0.25–16.63 $\mu\text{g/g}$ in captive samples vs. $<1.60 \mu\text{g/g}$ in wild samples) in captive samples were significantly greater than those in the wild ones. In contrary, the contents of YLGYLEQLLR in the captive samples (0.52–27.17 $\mu\text{g/g}$) were significantly lower than those wild samples (0.02–0.09 $\mu\text{g/g}$). Moreover, except for W9 sample (2.72 $\mu\text{g/g}$), trace distributions ($<0.82 \mu\text{g/g}$) occurred for DVDAAYMNK in most batches.

4. Conclusion

In current study, attempts were devoted to characterize the tryptic peptidome in musk and to develop a method permitting the quantitative

analysis of primary tryptic peptides as well. NanoLC–Orbitrap MS was utilized for data acquisition of shotgun proteomics, and Human Swiss-Prot FASTA database afterwards facilitated amino acid sequence annotation for a total of 733 tryptic peptides that were sourced from 79 proteins. Keratins were the primary sources of those tryptic peptides. Eight abundant peptides were selected as the analytes for quantitative evaluation, and their quantitative MS parameters, such as ion transitions and CEs, were efficiently optimized using online ER-MS rather than Skyline software. After diverse method validation assays, the developed LC–SRM method was applied for the simultaneous determination of eight tryptic peptides in 28 batches of musk samples, including captive (C1–C18) and wild ones (W1–W10). Significant differences took place for the content patterns of concerned tryptic peptides between the captive and wild musk samples. Captive samples were rich of QSLEASLAETEGR, TLLDIDNTR, and EVATNSSELVQSGK, whereas wild samples were able to accumulate YLGYLEQLLR. Above all, LC–MS/MS enabled simultaneous determination of eight tryptic peptides in musk, indicating a feasible analytical tool for comprehensive quality evaluation of musk, as well as other protein-enriched animal medicines.

CRedit authorship contribution statement

Wenjing Liu: Methodology, Investigation, Writing - original draft. **Juan Yu:** Investigation, Formal analysis. **Wei Li:** Investigation, Formal analysis. **Zhenzhen Jiang:** Validation, Visualization. **Ting Li:** Validation, Visualization. **Libo Cao:** Data curation, Writing - original draft. **Pengfei Tu:** Conceptualization, Project administration. **Jun Li:** Writing - review & editing, Supervision. **Yuelin Song:** Conceptualization, Writing - review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jchromb.2021.122624>.

References

- [1] X.M. Liu, H.Q. Li, X. Xiao, D.Q. Yu, Studies on the anti-inflammatory protein of musk. I. isolation, purification and properties, *Dong Wu Xue Bao* 38 (1992) 302–307.
- [2] X.Y. Zhu, W.J. Wang, G.F. Xu, Y.L. Yang, L.M. Xue, The pharmacological activities of musk. II. The anti-inflammatory activity of the active components of musk, *Yao Xue Xue Bao* 23 (1988) 406–410.
- [3] W.J. Wang, M. Zhong, Y. Guo, L.E. Zhou, G.F. Cheng, X.Y. Zhu, Effects of musk glucoprotein on chemotaxis of polymorphonuclear leukocytes in vivo and in vitro, *Zhongguo Zhong Yao Za Zhi* 28 (2003) 59–62.
- [4] X. Li, F. Shi, L.P. Gong, B.J. Hang, D.Y. Li, L.L. Chi, Species-specific identification of collagen components in Colla corii asini using a nano-liquid chromatography tandem mass spectrometry proteomics approach, *Int. J. Nanomed.* 12 (2017) 4443–4454, <https://doi.org/10.2147/IJN.S136819>.
- [5] X.T. Ma, J.K. Zhang, J.Z. Liang, X.L. Ma, R.R. Xing, J.X. Han, L.H. Guo, Y. Chen, Authentication of Edible Bird's Nest (EBN) and its adulterants by integration of shotgun proteomics and scheduled multiple reaction monitoring (MRM) based on

- tandem mass spectrometry, *Food Res. Int.* 125 (2019), 108639, <https://doi.org/10.1016/j.foodres.2019.108639>.
- [6] V. Lange, P. Picotti, B. Domon, R. Aebersold, Selected reaction monitoring for quantitative proteomics: a tutorial, *Mol. Syst. Biol.* 4 (2008) 222, <https://doi.org/10.1038/msb.2008.61>.
 - [7] V. Vidova, Z. Spacil, A review on mass spectrometry-based quantitative proteomics: Targeted and data independent acquisition, *Anal. Chim. Acta* 964 (2017) 7–23, <https://doi.org/10.1016/j.aca.2017.01.059>.
 - [8] J. Chen, X.L. Cheng, F. Wei, Q.Q. Zhang, M.H. Li, S.C. Ma, Detection of gelatin adulteration in traditional chinese medicine: analysis of deer-horn glue by rapid-resolution liquid chromatography-triple quadrupole mass spectrometry, *J. Anal. Methods Chem.* 2015 (2015), 259757, <https://doi.org/10.1155/2015/259757>.
 - [9] S.W. Guo, X.B. Xu, X.S. Zhou, Y.Q. Huang, A rapid and simple UPLC-MS/MS method using collagen marker peptides for identification of porcine gelatin, *RSC Adv.* 8 (2018) 3768–3773, <https://doi.org/10.1039/C7RA12539A>.
 - [10] R. Liu, Y. Huang, H.K. Xu, Y.F. Zheng, Y. Liu, S.Y. Han, M. Yang, Y. Xie, K.X. Wang, J.A. Duan, L.J. Li, A strategy for identifying species-specific peptide biomarkers in deer-hide gelatin using untargeted and targeted mass spectrometry approaches, *Anal. Chim. Acta* 1092 (2019) 32–41, <https://doi.org/10.1016/j.aca.2019.09.064>.
 - [11] A.J. Kleinnijenhuis, F.L. van Holthoorn, G. Herregods, Validation and theoretical justification of an LC-MS method for the animal species specific detection of gelatin, *Food Chem.* 243 (2018) 461–467, <https://doi.org/10.1016/j.foodchem.2017.09.104>.
 - [12] H.D. Yin, J.H. Zhu, M.M. Wang, Z.P. Yao, D.M. Lubman, Quantitative analysis of α -1-antitrypsin glycosylation isoforms in HCC patients using LC-HCD-PRM-MS, *Anal. Chem.* 92 (2020) 8201–8208, <https://doi.org/10.1021/acs.analchem.0c00420>.
 - [13] Z.A. Chen, J. Rappsilber, Quantitative cross-linking/mass spectrometry to elucidate structural changes in proteins and their complexes, *Nat. Protoc.* 14 (2019) 171–201, <https://doi.org/10.1038/s41596-018-0089-3>.
 - [14] X. Xu, W. Li, T. Li, K. Zhang, Q.Q. Song, L. Liu, P.F. Tu, Y.T. Wang, Y.L. Song, J. Li, Direct infusion-three-dimensional-mass spectrometry enables rapid chemome comparison among herbal medicines, *Anal. Chem.* 92 (2020) 7646–7656, <https://doi.org/10.1021/acs.analchem.0c00483>.
 - [15] W.J. Liu, Y. Cao, Y. Ren, X. Xu, L.Q. He, R.X. Xia, P.F. Tu, Y.T. Wang, Y.L. Song, J. Li, Simultaneously quantitative analysis of peptides and chemical components in *Cervus* and *Cucumis* polypeptide injection (*Songmeile*®) using reversed phase liquid chromatography-hydrophilic interaction liquid chromatography-tandem mass spectrometry, *J. Chromatogr. A* 1617 (2020), 460827, <https://doi.org/10.1016/j.chroma.2019.460827>.
 - [16] Q.Q. Song, J. Li, H.X. Huo, Y. Cao, Y.T. Wang, Y.L. Song, P.F. Tu, Retention time and optimal collision energy advance structural annotation relied on LC-MS/MS: an application in metabolite identification of an antidiabetic agent namely echinacoside, *Anal. Chem.* 91 (2019) 15040–15048, <https://doi.org/10.1021/acs.analchem.9b03720>.
 - [17] Y. Cao, C.C. Chai, A.Q. Chang, X. Xu, Q.Q. Song, W.J. Liu, J. Li, Y.L. Song, P.F. Tu, Optimal collision energy is an eligible molecular descriptor to boost structural annotation: An application for chlorogenic acid derivatives-focused chemical profiling, *J. Chromatogr. A* 1609 (2020), 460515, <https://doi.org/10.1016/j.chroma.2019.460515>.
 - [18] W.J. Liu, Q.Q. Song, Y. Cao, Y.F. Zhao, H.X. Huo, Y.T. Wang, Y.L. Song, J. Li, P. F. Tu, Advanced liquid chromatography-mass spectrometry enables merging widely targeted metabolomics and proteomics, *Anal. Chim. Acta* 1069 (2019) 89–97, <https://doi.org/10.1016/j.aca.2019.04.013>.
 - [19] The state pharmacopoeia commission of P. R. China, Pharmacopoeia of the People's Republic of China, English edition 2020, Beijing: China Medical Science Press, 402.
 - [20] Y. He, J.Z. Wang, M.Z. Wang, J.F. Zhang, Discrimination of wild and domestic deer musk using isotope ratio mass spectrometry, *J. Mass Spectrom.* 53 (2018) 1078–1085, <https://doi.org/10.1002/jms.4283>.
 - [21] R. Sun, Q. Yang, J.W. Yin, L.N. Zhang, L.L. Zhu, Research progress on the pharmacological action and determination method of moschus and substitutes, *Shizhen Guoyi Guoyao* 22 (2011), 709–710+712.
 - [22] W.I. Kimura, M. Ishida, Fundamental research for the pharmacological activity of oriental drugs. XI. combined effects of animal-origin drugs on the inhibition of leucocyte migration and their effective components, *Yakugaku Zasshi* 98 (1978) 442–447.
 - [23] X.M. Liu, X. Xiao, D.Q. Yu, Comparative study on some physicochemical properties of anti-inflammatory polypeptide proteins from captive and natural musk, *Zhong Cao Yao* 24 (1993) 10–12.
 - [24] D.Q. Yu, X.M. Liu, S.Y. Gao, Study on anti-inflammatory active ingredients of natural musk, *Yao Xue Xue Bao* 15 (1980) 306–307.
 - [25] R. Kanlaya, V. Thongboonkerd, Quantitative peptidomics of endogenous peptides involved in TGF- β 1-induced epithelial mesenchymal transition of renal epithelial cells, *Cell Death Discov.* 4 (2018) 9, <https://doi.org/10.1038/s41420-017-0001-x>.
 - [26] A. Bairoch, R. Apweiler, The SWISS-PROT protein sequence data bank and its new supplement TREMBL, *Nucleic Acids Res.* 24 (1996) 21–25, <https://doi.org/10.1093/nar/24.1.21>.
 - [27] Y.L. Song, Q.Q. Song, J. Li, J. Zheng, C. Li, Y. Zhang, L.L. Zhang, Y. Jiang, P.F. Tu, An integrated platform for directly widely-targeted quantitative analysis of feces part II: An application for steroids, eicosanoids, and porphyrins profiling, *J. Chromatogr. A* 1460 (2016) 74–83, <https://doi.org/10.1016/j.chroma.2016.07.008>.
 - [28] C.A. Holstein Sherwood, P.R. Gaffken, D.B. Martin, Collision energy optimization of b- and y-ions for multiple reaction monitoring mass spectrometry, *J. Proteome Res.* 10 (2011) 231–240, <https://doi.org/10.1021/pr1004289>.
 - [29] B. Maclean, D.M. Tomazela, S.E. Abbatiello, S. Zhang, J.R. Whiteaker, A. G. Paulovich, S.A. Carr, M.J. MacCoss, Effect of collision energy optimization on the measurement of peptides by selected reaction monitoring (SRM) mass spectrometry, *Anal. Chem.* 82 (2010) 10116–10124, <https://doi.org/10.1021/ac102179j>.
 - [30] C.A. Sherwood, A. Eastham, L.W. Lee, J. Risler, H. Mirzaei, J.A. Falkner, D. B. Martin, Rapid optimization of MRM-MS instrument parameters by subtle alteration of precursor and product m/z targets, *J. Proteome Res.* 8 (2009) 3746–3751, <https://doi.org/10.1021/pr801122b>.
 - [31] B. MacLean, D.M. Tomazela, N. Shulman, M. Chambers, G.L. Finney, B. Frewen, R. Kern, D.L. Tabb, D.C. Liebler, M.J. MacCoss, Skyline: an open source document editor for creating and analyzing targeted proteomics experiments, *Bioinformatics* 26 (2010) 966–968, <https://doi.org/10.1093/bioinformatics/btq054>.
 - [32] J.D. Egerton, B. MacLean, R. Johnson, Y. Xuan, M.J. MacCoss, Multiplexed peptide analysis using data-independent acquisition and Skyline, *Nat. Protoc.* 10 (2015) 887–903, <https://doi.org/10.1038/nprot.2015.055>.
 - [33] L.K. Pino, B.C. Searle, J.G. Bollinger, B. Nunn, B. MacLean, M.J. MacCoss, The Skyline ecosystem: Informatics for quantitative mass spectrometry proteomics, *Mass Spectrom. Rev.* 39 (2020) 229–244, <https://doi.org/10.1002/mas.21540>.
 - [34] Y. Yan, Q.Q. Song, X.J. Chen, J. Li, P. Li, Y.T. Wang, T.X. Liu, Y.L. Song, P.F. Tu, Simultaneous determination of components with wide polarity and content ranges in *Cistanche tubulosa* using serially coupled reverse phase-hydrophilic interaction chromatography-tandem mass spectrometry, *J. Chromatogr. A* 1501 (2017) 39–50, <https://doi.org/10.1016/j.chroma.2017.04.034>.
 - [35] Y.L. Song, N. Zhang, Y. Jiang, J. Li, Y.F. Zhao, S.P. Shi, P.F. Tu, Simultaneous determination of aconite alkaloids and ginsenosides using online solid phase extraction hyphenated with polarity switching ultra-high performance liquid chromatography coupled with tandem mass spectrometry, *RSC Adv.* 5 (2015) 6419–6428, <https://doi.org/10.1039/C4RA14088E>.
 - [36] Y.L. Song, W.H. Jing, G. Du, F.Q. Yang, R. Yan, Y.T. Wang, Qualitative analysis and enantiospecific determination of angular-type pyranocoumarins in *Peucedani Radix* using achiral and chiral liquid chromatography coupled with tandem mass spectrometry, *J. Chromatogr. A* 1338 (2014) 24–37, <https://doi.org/10.1016/j.chroma.2014.01.078>.
 - [37] K. Zhang, W.J. Liu, Q.Q. Song, J.B. Wan, J. Yu, X.C. Gong, L.B. Cao, D.D. Si, P. F. Tu, J. Li, Y.L. Song, Integrated strategy drives direct infusion-tandem mass spectrometry as an eligible tool for shotgun pseudo-targeted metabolomics of medicinal plants, *Anal. Chem.* 93 (2021) 2541–2550, <https://doi.org/10.1021/acs.analchem.0c04602>.